

Green House Gas (GHG) Emission Report 2024

About the Report

The Granules GHG Emissions Report provides a comprehensive overview of the **Granules India Limited (GIL) and subsidiaries (GPI ,GUSA ,GLS ,GCZRO)** greenhouse gas emissions, covering Scope 1, Scope 2, and Scope 3 categories.

As part of our commitment to reducing environmental impact, we have analyzed emissions directly under our control (Scope 1), such as fuel combustion in company-owned facilities and vehicles, as well as indirect emissions from purchased electricity, steam, and heating (Scope 2).

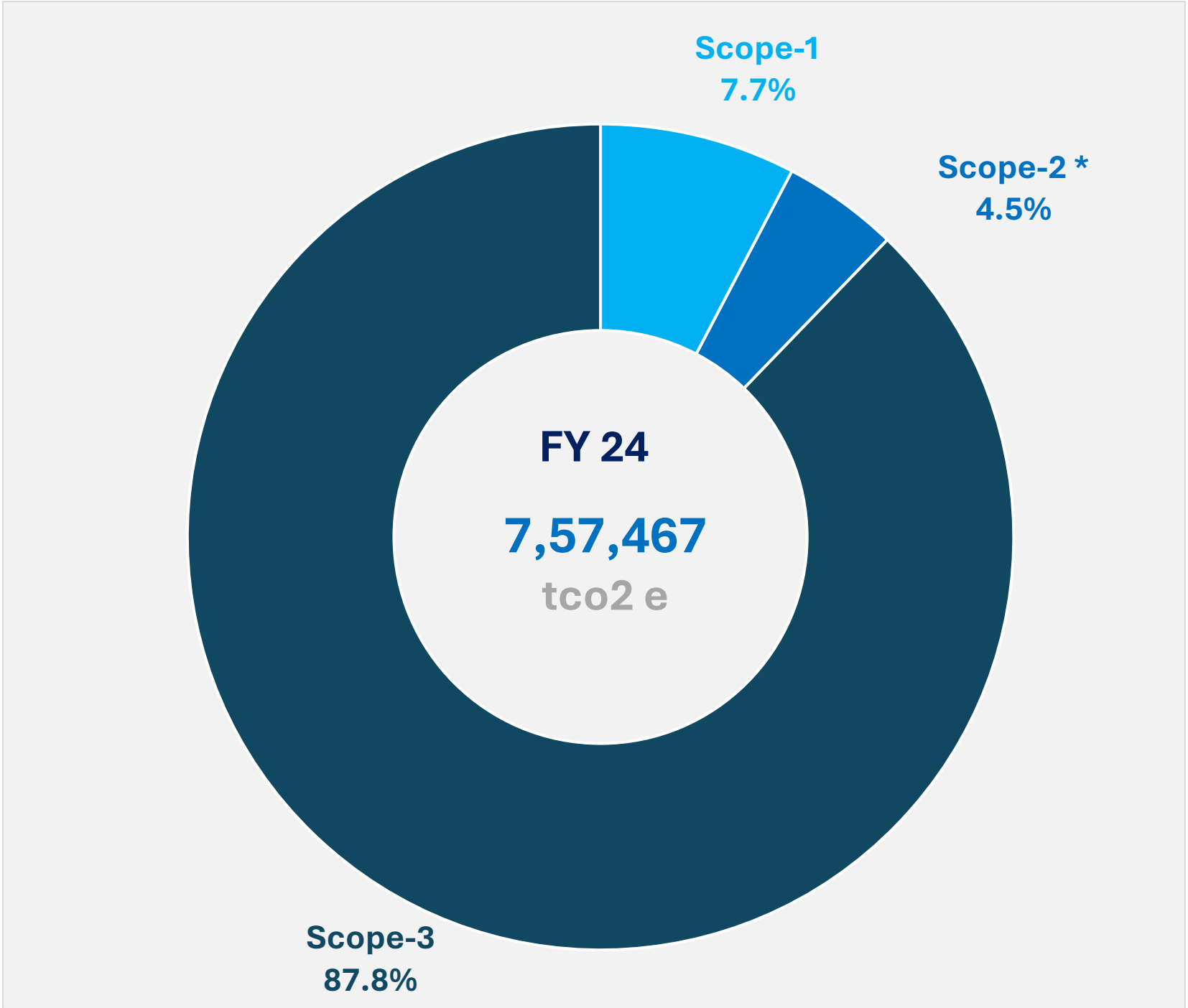
And Scope 3 emissions are accounted for, encompassing emissions across our entire value chain, from supplier activities to product use and disposal. Recognizing that Scope 3 emissions often represent the largest portion of our carbon footprint, we are working to collaborate with suppliers, optimize operations, and engage stakeholders throughout our supply chain.

This report reflects our dedication to transparency and our ongoing efforts to meet industry standards, particularly aligning with the GHG Protocol, while advancing our sustainability initiatives to mitigate climate impact.

Our Carbon Footprint – Overview

Scope	FY 23	FY 24
Scope-1	57,816	58,254
Scope-2 (Market Based)	57,974	34,519
Scope-3	6,25,236	6,66,694
Total Emissions	7,41,026	7,59,467

All Values in Tons of Co₂ equivalent

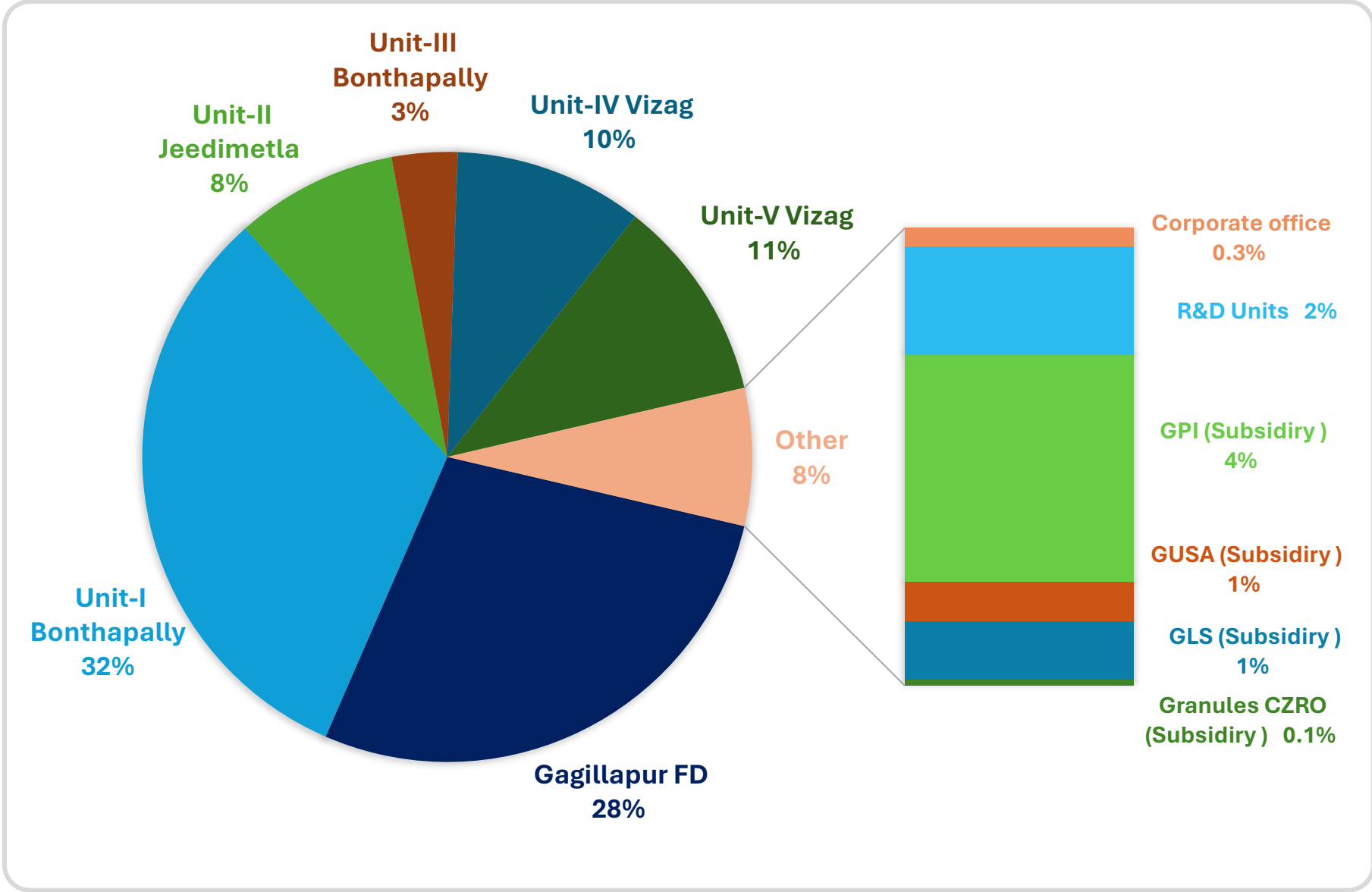


*Market –Based

Our Carbon Footprint – Detailed view

Granules GHG Emission Profile , tCo2 eq		Emission Profile For FY 2022-23				Emission Profile For FY 2023-24		
Scope	Emission Category	GIL Standalone	GIL-All Subsidiaries	GIL Consolidated		GIL Standalone	GIL-All Subsidiaries	GIL Consolidated
Scope-1	Direct emissions from sources owned or controlled by a company (Coal, Fuels)	56,265	1,551	57,816		56,254	2,001	58,254
Scope-2 (Market-Based)	Indirect emissions from the generation of purchased energy (Electricity)	56,306	1,668	57,974		31,632	2,886	34,519
Scope-2 (Location -Based)	Indirect emissions from the generation of purchased energy (Electricity)	62,433	1,668	64,101		60,191	2,886	63,078
	Scope-1+2 Total	1,12,571	3,219	1,15,790		87,886	4,887	92,773
Scope-3	Category 1 - Purchased Goods & Services	4,95,807	882	4,96,689		5,14,516	5,292	5,19,808
Scope-3	Category 2 - Capital Goods	4,705	465	5,170		10,265	18,565	28,830
Scope-3	Category 3 - Fuel and Energy Related Activities	26,788	403	27,191		22,579	875	23,453
Scope-3	Category 4 - Upstream Transportation & Distribution	8,392	33	8,425		8,557	83	8,640
Scope-3	Category 5 - Waste Generated from Operations	2,992	1,625	4,618		3,571	150	3,721
Scope-3	Category 6 - Business Travel	295	37	332		850	81	931
Scope-3	Category 7 - Employee Commuting	1,476	314	1,790		1,571	518	2,089
Scope-3	Category 8 - Upstream Leased Assets	587	4,205	4,792		1,628	6,153	7,781
Scope-3	Category 9 - Downstream Transportation & Distribution	22,381	3,131	25,512		24,827	1,097	25,924
Scope-3	Category 10 - Processing of Sold Products	46,658	-	46,658		41,187	1	41,188
Scope-3	Category 11 - Use of Sold Products	-	-	-		-	-	-
Scope-3	Category 12 - End of Life Treatment of sold products	4,031	28	4,059		4,287	43	4,330
Scope-3	Category 13 - Downstream Leased Assets	-	-	-		-	-	-
Scope-3	Category 14 - Franchises	-	-	-		-	-	-
Scope-3	Category 15 - Investments	-	-	-		-	-	-
	Scope-3 Total	6,14,113	11,123	6,25,236		6,33,837	32,857	6,66,694
Total Emissions - Scope 1 ,Scope 2 (marked Based) and Scope 3		7,26,684	14,342	7,41,026		7,21,723	37,744	7,59,467

Scope 1 and Scope 2 – Analysis by site (FY 23-24)

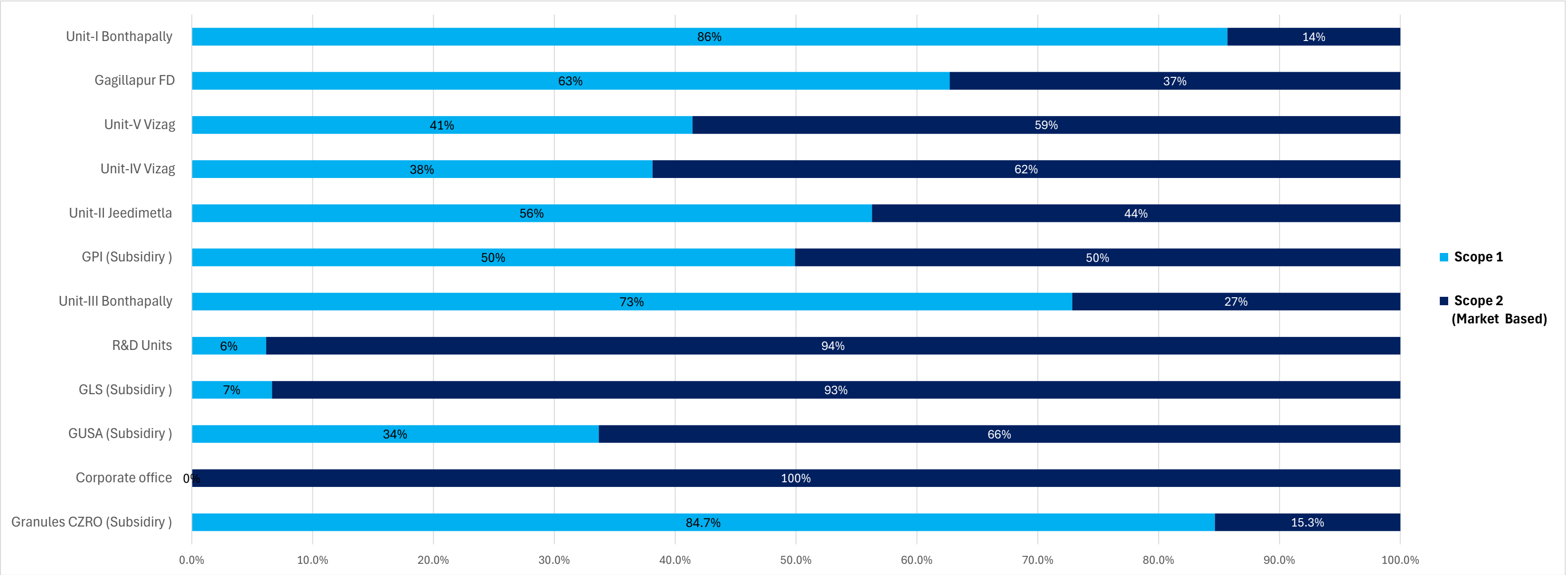


Unit & Location	Scope 1 tCo2e	Scope 2 , tco2e (Market Based)	Total (Scope 1 +Scope 2 Market based)
Gagillapur FD	16,227	9,644	25,871
Unit-I Bonthapally	25,402	4,240	29,641
Unit-II Jeedimetla	4,484	3,482	7,966
Unit-III Bonthapally	2,351	876	3,227
Unit-IV Vizag	3,540	5,746	9,286
Unit-V Vizag	4,152	5,867	10,019
Corporate office	-	272	272
R&D Units	99	1,504	1,603
GPI (Subsidiary)	1,673	1,679	3,352
GUSA (Subsidiary)	196	386	582
GLS (Subsidiary)	57	808	866
Granules CZRO (Subsidiary)	74	13	87
Total - Scope 1 +Scope 2 (Market Based) , tCo2 e	58,254	34,519	92,773

Scope2 as per market-based method

The market-based method reflects the renewable adoption through power purchase agreement (PPA), solar roof top at site and purchase of Energy Attribute Certificates (EAC) through International Renewable Energy Certificates (IREC).

Scope 1 and Scope 2 – Analysis by site (FY 23-24)

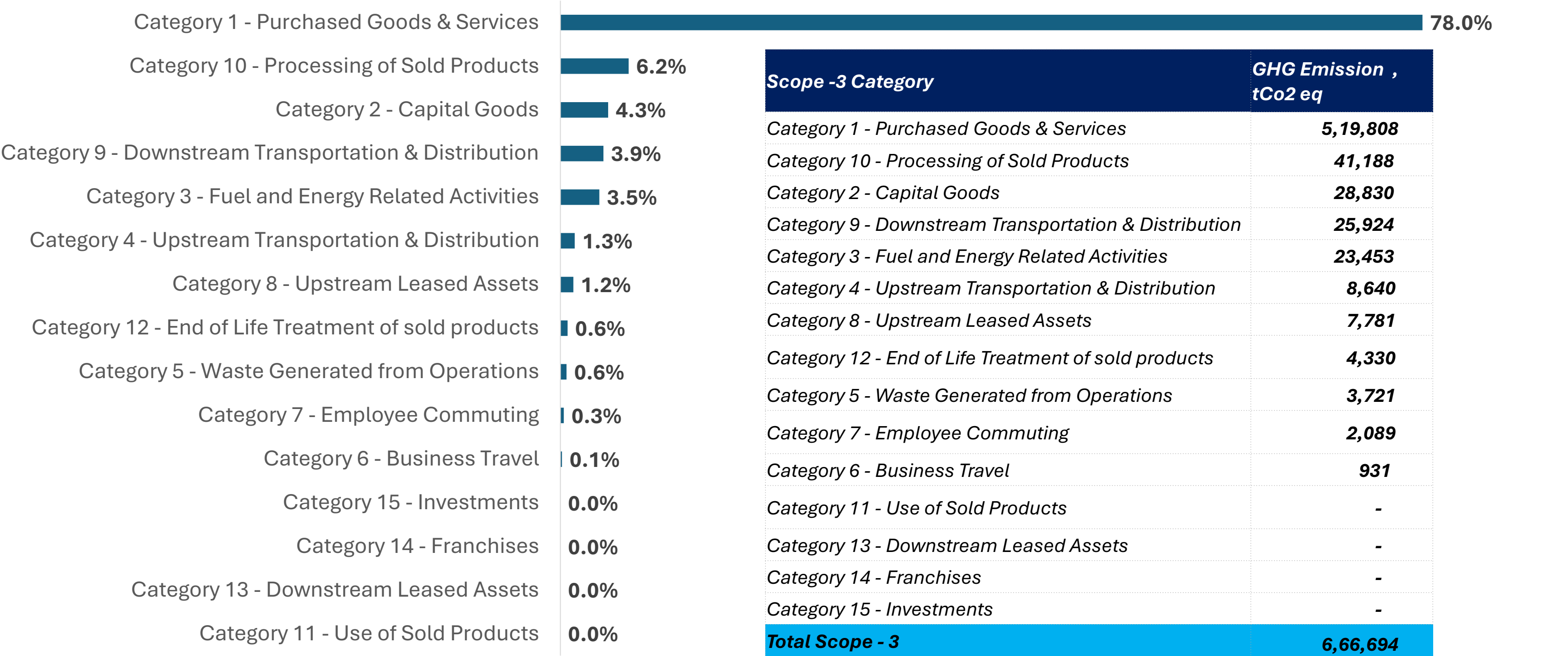


Unit wise Profile -Scope1 and Scope 2 Emissions (Marked Based Method)

Scope2 as per market-based method

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Scope 3 – Analysis by Category (FY 23-24)



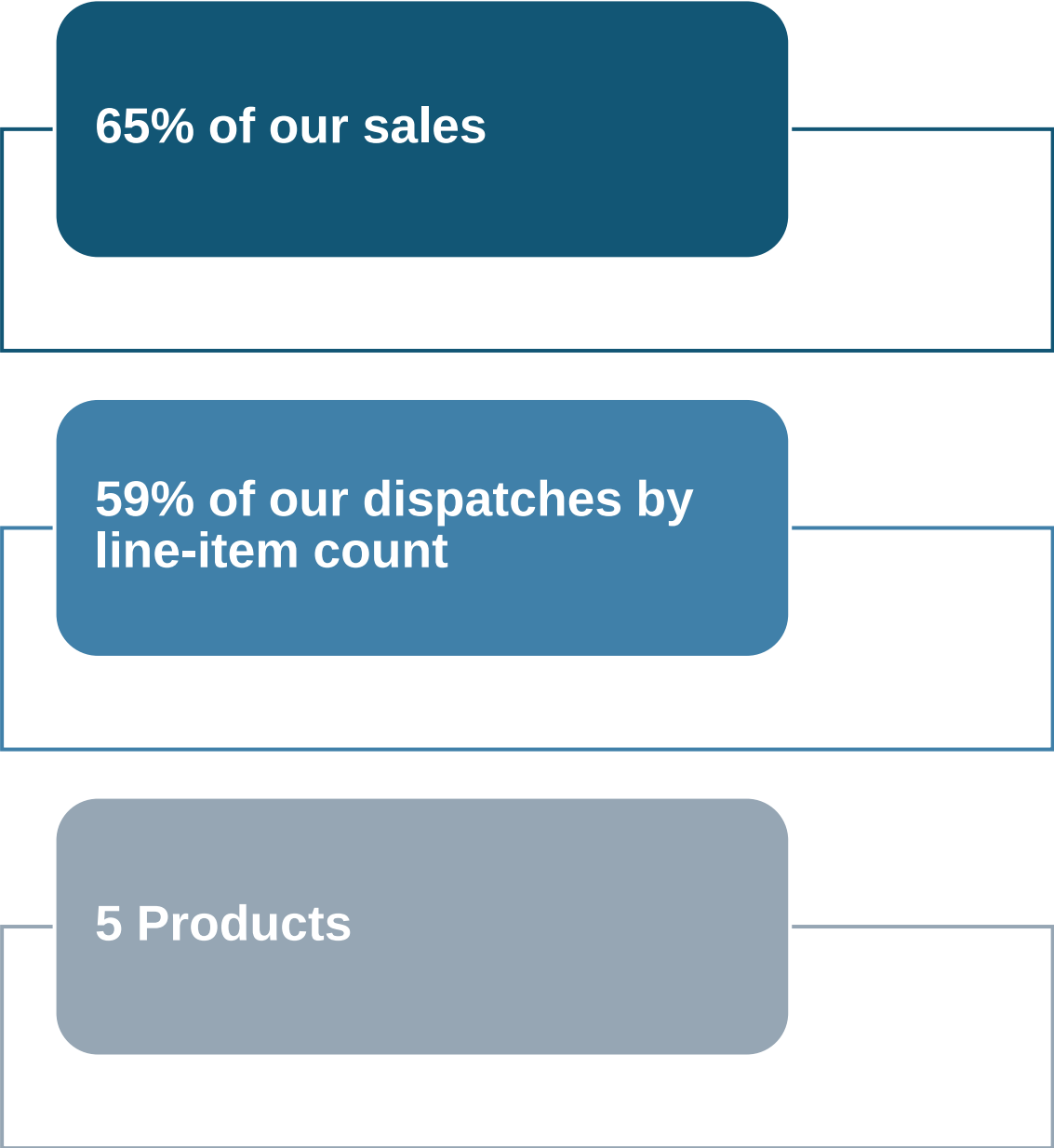


Our Journey

Carbon Footprint to Carbon – Free

Product Carbon Footprint

We have estimated the Product Carbon Footprint (PCF) for 5 Key molecules (for all SKUs) of our molecules covering 65% of our product sales. In the near future, we will conduct Life cycle assessments covering all the impact categories.



PCF Coverage – Product in Focus	
Product Choice <i>Product on which the GHG life cycle inventory is performed</i>	• Paracetamol (API / PFI / FD) • Metformin (API / PFI / FD) • Potassium Chloride (PFI / FD) • Metoprolol Succinate (API / PFI / FD) • Venlafaxine (PFI / FD)
Unit of Analysis <i>Performance characteristics and services delivered by the product being studied</i>	• Quality grade – 1st / 2nd crop (for API) • Strength / Concentration (for PFI / FD) • Dosage form - Tablet / Capsules / Gel Caps / Caplets (for FD) • Release profile - IR / ER (for FD)
Reference Flow <i>Amount of product on which the results of the study are based</i>	• MT (for API and PFI) • Number of tablets (for FD)
Coverage	• Cradle-to-Factory-Gate-to-Customer-Gate <i>Does not include Product Processing by customer, usage, end-of-life treatment etc.</i>

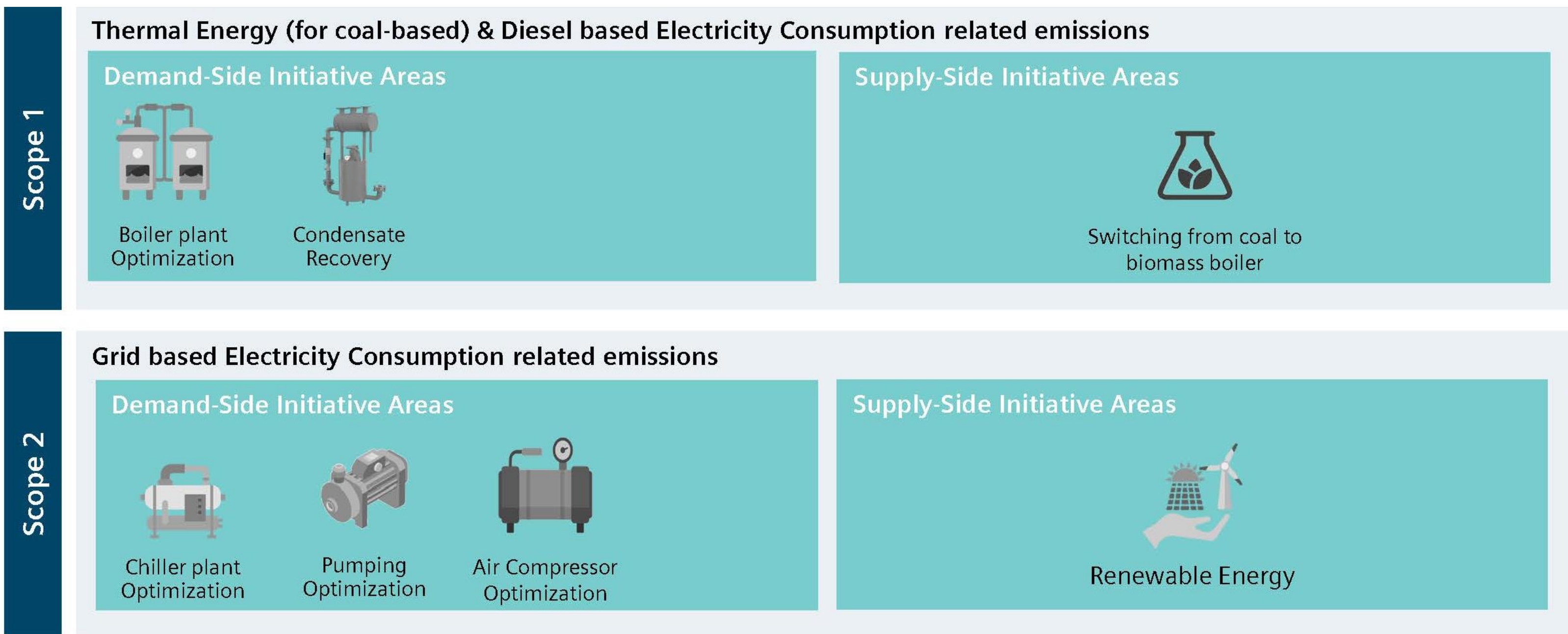


Decarbonization Action Plan

Our Decarbonization Action Plan:

Addressing Scope 1 and Scope 2 emissions

To reduce Scope 1 and Scope 2 GHG emissions a balanced approach of supply-side and demand-side initiatives has been identified to support decarbonization till 2030. It identifies that beyond 2030 a high green energy support will be needed to meet the decarbonization goals. This strategy is based on a detailed study at the Bonthapally and Gagillapur facilities, where the primary sources of emissions are coal-fired boilers and grid electricity. Granules will utilize a similar approach for other operational locations.



Our Decarbonization Action Plan:

Addressing Scope 3 emissions- Various levers identified

#	Category	Recommended initiatives	Carbon reduction impact	Implementation time	Feasibility
	Cat 1: Purchased Goods & Services	Supplier Mix Optimization	High	Medium	High
		Scope 1/2 Targets	High	Low	Medium
		Sustainable Packaging	Low	Medium	Low
	Cat 3: Fuel & Energy Related Activities	Emissions savings due to coal savings from FIMs	Medium	Low	High
		Emissions savings due to electricity savings from FIMs	Low	Low	High
		Emission savings due to grid decarbonization	High	High	High
	Cat 4: Upstream T&D	Consolidating Shipments			
		Utilizing Multi-Modal Transportation			
	Cat 5: Waste	Reduction of non-hazardous paper waste	Low	High	Medium
	Cat 6: Business Travel	Optimize Mode of Travel	Low	Low	High
		Green Travel Options	Low	Low	High
	Cat 7: Employee Commute	Flexible Working Hours	Low	Low	High
		Carpooling & Shared Mobility	Low	Low	High
		Offer EV charging Infra and Leasing Options for EV Adoption	Low	Medium	High
		EV Fleet for company buses / cars	Low	Medium	High
	Cat 8: Upstream Leased Assets	Integrate renewable energy and implement BMS	Low	Medium	Medium
	Cat 9: Downstream T&D	Transition of shipment mode from air to sea	Medium	High	Low
		Consolidation of container utilization	Low	High	Low
		Shift to EVs	Low	High	Low

Our Decarbonization Action Plan:

Pioneering Green Science (3-Pillar approach)

Green Molecule Platform



GRANULES

CZRO

Building our flagship green pharmaceutical plant

Green molecules

Such as Hydrogen (H_2), Ammonia (NH_3), Nitric acid (HNO_3) and Methanol (CH_3OH)



Built-in circularity

Green Chemistry



Granules eco-scale for product development



Enzyme and bio transformation for resource-efficient processes



Flow technology and continuous manufacturing platform

Sustainable Formulations



Plant-based excipients



Solvent/Polymer-free formulations



Water-efficient manufacturing



Biodegradable & sustainable packing

Our Decarbonization Action Plan:

Supplier Sustainability Program

GHG Reduction Targets	Encourage suppliers to set and commit to measurable GHG reduction targets aligned with industry standards.
Data Sharing and Transparency	Promote regular reporting and data sharing on GHG emissions, ensuring transparency and accountability throughout the supply chain
Training and Capacity Building:	Provide resources and training to help suppliers understand GHG reduction practices and implement energy-efficient solutions
Collaboration on Innovation	Partner with suppliers to co-develop low-carbon technologies, materials, and processes to reduce emissions across operations.
Incentives and Recognition	Offer incentives or recognition programs for suppliers that demonstrate significant progress in their GHG reduction efforts.



Our Supplier Engagement Approach

Our Decarbonization Action Plan:

Supplier Sustainability Program

Our Ask to Supplier Commitment



1. Disclose Carbon Footprint
Disclose Scope 1, 2, and 3 emissions related to Granules business by the end of 2024.



2. Commitment to Science Based Target Initiative (SBTi)
Develop and submit a science-based target to the SBTi by the end of 2025.



3. Declare Product Carbon Footprint (PCF)
Provide Product carbon footprint (PCF) for products and facilities for business with Granules.



4. Commit to Accelerating Renewable Energy Adoption Increase renewable energy in the electricity purchase and to 100% renewable energy within next 5 years.

We have rolled out the supplier sustainability program for top suppliers covering the more than 80 % of our purchase value.



Granules India Limited

2nd Floor | 3rd Block | My Home Hub Madhapur

Hyderabad | India | 500 081

Tel: +91-40-6904 3500 |

<https://granulesindia.com/>